JUTHELINK · PHASE 1

Set up your MikroTik hotspot and Tenda outdoor access point

A complete, first-time, click-by-click walkthrough. At the end, customers connect to “Starling Wi-Fi”, see your login page, type the username and password you sold them, and are disconnected automatically when their time is up.

Your equipment
Starlink · MikroTik hAP lite (RouterOS 6.48.6) · Tenda outdoor AP

Time needed
About 2–3 hours, with breaks. Do not rush.

Skill needed
None. Every click and every command is written here.

0 · Before you start 2
1 · Winbox basics 4
2 · Start clean 5
3 · Password, name, clock 5
4 · Internet from Starlink 7
5 · Update the router 8
6 · Customer network (bridge) 9
7 · The router's own Wi-Fi (for testing) 10
8 · Turn on the hotspot (captive portal) 11
9 · Let your own PC/phone skip the login 12
10 · Hotspot settings 13
11 · Packages and prices (profiles) 14
12 · Your login page 16
13 · How selling works (MoMo + WhatsApp) 18
14 · Tenda outdoor access point 19
15 · Final test checklist 21
16 · Daily use 22
17 · Problems and fixes 23
A · All commands in one place 24

Tip: This guide sets up the router to work **on its own** (phase 1): you create usernames and passwords, sell them for cash or Mobile Money, and the router does the rest. Later, the Starling website will be connected to it (phase 2). Nothing in this guide needs to be redone for that.

PART 0

Before you start

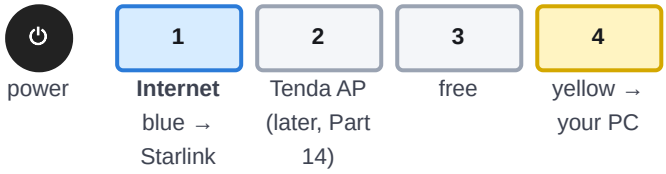
What you are building, what you need, and three rules that keep you safe.

What you are building

Starlink → blue cable → **MikroTik port 1** … MikroTik … **port 2** → **Tenda AP** → **customer phones**

The MikroTik sits in the middle. Internet comes in on **port 1**. Everything plugged into **ports 2, 3, 4** (and its own Wi-Fi) is the customer side: anyone there must log in on your page before they get internet.

Your cables (from your photo)



Looking at the back of the hAP lite. Port 1 is the one next to the power socket, marked “Internet”. Keep it like this for the whole guide.

What you need

- [ ] Starlink set up and working (the Starlink app shows **Online**).
- [ ] The MikroTik hAP lite, powered, blue cable in port 1, yellow cable in port 4 to your PC.
- [ ] Your PC with **Winbox** (you already have it and can connect by MAC address).
- [ ] A phone for testing.
- [ ] The kit `Starling-Router-Setup-Kit.zip` (unzip it on your PC): `starling-profiles.rsc`, `ticket-maker.html`, and the `hotspot` folder with `login.html`, `alogin.html`, `status.html`.
- [ ] The Tenda outdoor AP, its PoE injector and power adapter (only for Part 14).
- [ ] A notebook to write down passwords.

Three rules

Careful: Always connect Winbox using the MAC address (the letters/numbers like `18:FD:74:FF:17:EB`), not an IP address, until Part 9. The MAC connection keeps working even if you make a mistake with IP settings.

Careful: Type values exactly as written. Small letters stay small. No extra spaces. When this guide says `bridge-hotspot`, type exactly that.

Careful: If you get lost, you can always start again with Part 2 (start clean). Nothing you do here can damage the router.

Write your values here

What	Value
Router MAC address (Winbox)	
Router admin password (Part 3)	
Wi-Fi name (SSID)	Starling Wi-Fi
Hotspot admin test user / password (Part 8)	
Your WhatsApp / MoMo number (Part 12)	
Tenda admin password (Part 14)	

PART 1

Winbox basics

Five things you will use in every step.

1 Connecting

Open Winbox → **Neighbors** tab → **Refresh** → click the line that shows **hAP lite** → make sure the **MAC address** appears in “Connect To” → Login `admin`, Password: empty (or the password you set in Part 3) → **Connect**.

2 The menu on the left

Everything is in the left menu: **Interfaces**, **Wireless**, **Bridge**, **IP** (has a sub-menu: Addresses, DHCP Client, DNS, Hotspot, Firewall…), **System** (sub-menu: Password, Identity, Clock, Packages, Reset Configuration…), **Files**, **New Terminal**. In this guide, “**IP → DNS**” means: click IP, then DNS in the sub-menu.

3 Adding and editing

In most windows, the blue **+** (or **New**) button adds an item. Double-click a line to edit it. In an edit window, **Apply** saves and keeps it open, **OK** saves and closes.

4 The terminal (for copy-paste commands)

New Terminal opens a black window. You can paste a command there (right-click → Paste, or Ctrl+V) and press **Enter**. Every step in this guide shows both ways: the clicks, and the command. **Use one or the other for a step, not both.** If you are unsure, use the clicks.

5 Safe Mode (your undo button)

The **Safe Mode** button (top right in Winbox) protects you: while it is on, if Winbox loses the connection, the router automatically **undoes** your last changes. Turn it on before risky steps (this guide tells you when), and turn it off again when you are happy with the result so the changes are kept.

Tip: Your Winbox may look slightly different (newer Winbox 4 has a darker style and some buttons are named “New” instead of “+”). The menu names are the same.

PART 2

Start clean

Your router is almost empty already (it showed a `bridge1`). We erase everything so it is in a known state.

1. **Reset with no configuration**

   System → Reset Configuration → tick **No Default Configuration** → leave the other boxes unticked → **Reset Configuration** → confirm **Yes**.

   ```
   /system reset-configuration no-defaults=yes skip-backup=yes
   ```

   (In the terminal, type `y` and Enter when it asks “Dangerous! Reset anyway?”)

2. **Wait and reconnect**

   Winbox disconnects. Wait about 1 minute (the router reboots). Then Winbox → Neighbors → Refresh → click the MAC address → login `admin`, empty password → Connect. If it asks about the default configuration, click **Remove Configuration** (or OK).

✔ **You should see: Bridge** is empty, **IP → Addresses** is empty. That is correct.

✖ **If not:** The router does not appear in Neighbors: unplug the yellow cable from the PC and plug it back in, wait 20 seconds, Refresh again. Try another port (2 or 3) on the router.

PART 3

Password, name and clock

1. **Set the admin password**

   System → Password → Old Password: empty → New Password and Confirm: a strong password (write it in Part 0) → **Change**.

   ```
   /password old-password="" new-password="YourStrongPassword" confirm-new-password="YourStrongPassword"
   ```

2. **Give the router a name**

   System → Identity → `starling-hotspot` → OK.

   ```
   /system identity set name=starling-hotspot
   ```

3. **Time zone and automatic clock**

   System → Clock → Time Zone Name: `Africa/Douala` → OK. Then **System → SNTP Client** → tick **Enabled** → Primary NTP Server `162.159.200.1`, Secondary `162.159.200.123` → OK.

   ```
   /system clock set time-zone-name=Africa/Douala
   /system ntp client set enabled=yes primary-ntp=162.159.200.1 secondary-ntp=162.159.200.123
   ```

   The clock matters: customers' time is counted with it. It will show the right time once the internet works (Part 4).

Tip: From now on, log in with `admin` and your new password.

PART 4

Internet from Starlink (port 1)

The router asks the Starlink router for an address on port 1, exactly like a phone does.

---

1 Check Starlink

The Starlink app says **Online**, and the blue cable goes from a Starlink router Ethernet/LAN port to **port 1** of the MikroTik.

2 Add a DHCP client on port 1

IP → DHCP Client → **+** → Interface: `ether1` → **Use Peer DNS: unticked** → Add Default Route: **yes** → OK.

```
/ip dhcp-client add interface=ether1 use-peer-dns=no add-default-route=yes disabled=no
comment="Internet from Starlink"
```

✔ **You should see:** In the DHCP Client list, the Status becomes **bound** and an IP address appears (for example `192.168.1.23/24`, or `100.x.x.x` if Starlink is in bypass mode).

✖ **If not:** Status stays “searching…”: check the blue cable is in **port 1** and in the Starlink router; restart the Starlink router; wait 1 minute. If Starlink uses the Ethernet Adapter, check it is connected.

3 Set the DNS servers

IP → DNS → Servers: `1.1.1.1` and (click the small arrow to add a line) `8.8.8.8` → tick **Allow Remote Requests** → OK.

```
/ip dns set servers=1.1.1.1,8.8.8.8 allow-remote-requests=yes
```

4 Test the internet

New Terminal and type:

```
/ping google.com count=4
```

✔ **You should see:** Four lines with a time like `time=40ms` and “received=4”.

✖ **If not:** “could not resolve”: redo step 3. “timeout”: the DHCP client is not bound; redo step 2.

5 Check the clock

System → Clock: the date and time are now correct (Douala time). If not, wait 2 minutes and look again.

PART 5

Update the router

Always update before configuring the hotspot, so an update does not change your settings later. Your router has 6.48.6; we move to the latest 6.49 “long-term” version.

1 Download and install

System → Packages → Check For Updates → Channel: **long-term** → wait until it shows the Latest Version (6.49.something) → **Download&Install**.

```
/system package update set channel=long-term
/system package update check-for-updates
/system package update install
```

✔ **You should see:** The router downloads, then reboots by itself (Winbox disconnects). Wait 2–3 minutes, reconnect by MAC. **System → Packages** shows the new version.

2 Update the RouterBOARD firmware

System → RouterBOARD → if “Current Firmware” and “Upgrade Firmware” are different, click **Upgrade** → OK. Then **System → Reboot** → Yes. Reconnect after 1 minute.

```
/system routerboard upgrade
/system reboot
```

Careful: Do not unplug the power while it is updating. If the update fails or does not show 6.49, that is fine: continue with 6.48.6. Everything in this guide works on both.

Careful: Stay on version 6 (long-term). Do not choose the “stable/7” channel: the hAP lite has very little storage and this guide is written for version 6.

PART 6

Customer network (the bridge)

A “bridge” joins ports 2, 3, 4 and the Wi-Fi into one customer network. Port 1 (internet) must NOT be in it.

1 Create the bridge

Bridge → **Bridge** tab → **+** → Name: `bridge-hotspot` → OK.

```
/interface bridge add name=bridge-hotspot comment="Customers"
```

2 Put ports 2, 3, 4 and the Wi-Fi in it

Bridge → **Ports** tab → **+** → Interface `ether2`, Bridge `bridge-hotspot` → OK. Repeat for `ether3`, `ether4` and `wlan1`.

```
/interface bridge port add bridge=bridge-hotspot interface=ether2
/interface bridge port add bridge=bridge-hotspot interface=ether3
/interface bridge port add bridge=bridge-hotspot interface=ether4
/interface bridge port add bridge=bridge-hotspot interface=wlan1
```

✔ **You should see:** The Ports tab lists ether2, ether3, ether4, wlan1, all with bridge **bridge-hotspot**. **ether1 is not there.**

Tip: Winbox may disconnect for a second because your PC (port 4) just moved into the bridge. Reconnect by MAC. This is normal.

3 Give the customer network its address

IP → **Addresses** → **+** → Address `10.5.50.1/24` → Interface `bridge-hotspot` → OK.

```
/ip address add address=10.5.50.1/24 interface=bridge-hotspot comment="Customer gateway"
```

✔ **You should see:** IP → Addresses shows `10.5.50.1/24` on bridge-hotspot (and the Starlink address on ether1, marked D).

Tip: 10.5.50.1 is your router's address for customers. Customers will get addresses 10.5.50.10 to 10.5.50.250. The Tenda will later get 10.5.50.2.

PART 7

The router's own Wi-Fi (for testing now)

The hAP lite has a small indoor Wi-Fi. We turn it on so you can test everything before mounting the Tenda. You can keep it on afterwards as a second, indoor access point.

---

1 **Make the Wi-Fi open (no password)**

The hotspot login page is the lock, so the Wi-Fi itself has no password (like airports and cafés).
Wireless → **Security Profiles** tab → double-click **default** → Mode: **none** → OK.

```
/interface wireless security-profiles set [find default=yes] mode=none
```

2 **Set the Wi-Fi name and turn it on**

Wireless → **WiFi Interfaces** tab → double-click **wlan1** → **Wireless** tab: Mode **ap bridge**, Band **2GHz-B/G/N**, Channel Width **20MHz**, Frequency **auto**, SSID `Starling Wi-Fi`, Security Profile **default** → OK. Then select wlan1 and click the blue tick ✓ **Enable**.

```
/interface wireless set wlan1 mode=ap-bridge band=2ghz-b/g/n channel-width=20mhz
frequency=auto ssid="Starling Wi-Fi" security-profile=default wireless-protocol=802.11
disabled=no
```

✔ **You should see:** On your phone, a Wi-Fi called **Starling Wi-Fi** appears, without a lock icon. Do not connect yet: there is no hotspot until Part 8, so the phone would not get an address.

PART 8

Turn on the hotspot (captive portal)

The Hotspot Setup wizard creates the login system, gives addresses to customers, and shares the internet. This is the same wizard as in the video.

1 Start the wizard

IP → Hotspot → **Servers** tab → **Hotspot Setup**. Answer each screen and click **Next**:

The wizard asks	You answer
HotSpot Interface	`bridge-hotspot`
Local Address of Network	`10.5.50.1/24  (Masquerade Network: ticked)`
Address Pool of Network	`10.5.50.10-10.5.50.250`
Select Certificate	`none`
IP Address of SMTP Server	`0.0.0.0`
DNS Servers	`1.1.1.1 and 8.8.8.8`
DNS Name	`login.starling`
Name of Local HotSpot User	`admin  + a password you choose (for your own tests; write it in Part 0)`

✔ **You should see:** "Setup has completed successfully". In the Servers tab there is one server on bridge-hotspot (Winbox names it **server1** or **hotspot1**; either is fine). **IP → DHCP Server** shows a server on bridge-hotspot. **IP → Firewall → NAT** shows a "masquerade hotspot network" rule.

Tip: The DNS Name `login.starling` is the address customers see at the top of the login page. It is not a real website; the router answers it itself.

2 Make sure FastTrack is not active

FastTrack is a speed shortcut that lets customers skip the login. You started clean, so there should be none, but check: **IP → Firewall → Filter Rules**. If any rule has Action **fasttrack connection**, select it and click the red **✗ Disable**.

```
/ip firewall filter disable [find action=fasttrack-connection]
```

3 First test with your phone

Phone → Wi-Fi → **Starling Wi-Fi** → connect. Open the browser and go to `http://neverssl.com`.

✔ **You should see:** MikroTik's grey login page appears (Part 12 replaces it with yours). Log in with `admin` and the hotspot password from step 1: you can browse.

✖ **If not:** **Phone says “Couldn't get IP address”** (and IP → DHCP Server → Leases stays empty): the Wi-Fi is not part of the customer bridge, or the wizard was run several times. Run this in New Terminal (safe to run more than once), then forget the network on the phone and join again:

```
/interface bridge port remove [find interface=wlan1]
/interface bridge port add bridge=bridge-hotspot interface=wlan1
/ip dhcp-server network remove [find]
/ip dhcp-server network add address=10.5.50.0/24 gateway=10.5.50.1 dns-server=10.5.50.1
/ip dhcp-server enable [find]
```

✖ **If not:** The phone says “no internet” and no page appears: type `http://login.starling` in the browser. Still nothing: check **IP → Hotspot → Servers**: hotspot1 must not be red (disabled). Check **IP → DHCP Server → Leases**: your phone must have a 10.5.50.x address.

PART 9

Let your own PC and phone skip the login

Your PC on port 4 is now on the customer network, so its browser would also need to log in. You can let your own devices through without a ticket.

1. **Find your PC's MAC address**

   IP → **DHCP Server** → **Leases** tab. Your PC is the line with your PC's name (Host Name). Note its **MAC Address**. (Also your phone's, from the test in Part 8.)

2. **Add a bypass**

   IP → **Hotspot** → **IP Bindings** tab → **+** → MAC Address: your PC's MAC → Type: **bypassed** → Comment: `Owner PC` → OK. Repeat for your own phone.

   ```
   /ip hotspot ip-binding add mac-address=AA:BB:CC:DD:EE:FF type=bypassed comment="Owner PC"
   ```

   (Replace AA:BB:CC:DD:EE:FF with the real MAC.)

✔ **You should see:** On your PC, any website opens without the login page.

Careful: Only add your own devices. Every bypassed device uses internet without paying and without speed limits.

PART 10

Hotspot settings

The settings the video recommends: one address per phone, and the login methods.

1 One address per phone

IP → **Hotspot** → **Servers** → double-click your server (**server1** or **hotspot1**) → Addresses Per MAC: `1` → OK.

```
/ip hotspot set [find] addresses-per-mac=1
```

2 Login methods

IP → **Hotspot** → **Server Profiles** tab → double-click **hsprof1** → **Login** tab → tick **HTTP CHAP**, **HTTP PAP**, **MAC Cookie**, **Cookie** (untick the others) → OK. (How long the MAC cookie lasts is set in the packages, Part 11.)

```
/ip hotspot profile set hsprof1 login-by=http-chap,http-pap,mac-cookie,cookie
```

Setting	Why
Addresses per MAC = 1	Each phone gets exactly one address.
HTTP PAP	Needed by your Starling login page (Part 12) and later the website's Connect button.
MAC Cookie	A phone that already logged in is let back in automatically after it reconnects, without typing again (until its time ends).

PART 11

Packages and prices (user profiles)

A “user profile” is a package: how long it lasts, how fast it is, how many phones can use it. Your six packages are ready in a file, including a small script that **starts the clock at the customer's first login and removes them automatically when the time is up**.

Profile	Length	Price	Phones	Speed (up/down)
`1hour`	1 hour	100 FCFA	1	2 / 5 Mbit/s
`12hours`	12 hours	150 FCFA	1	2 / 5 Mbit/s
`24hours`	24 hours	200 FCFA	1	2 / 5 Mbit/s
`1week`	7 days	1 350 FCFA	1	2 / 5 Mbit/s
`2weeks`	14 days	2 500 FCFA	1	2 / 5 Mbit/s
`1month`	30 days	4 500 FCFA	2	2 / 5 Mbit/s
`family`	never expires	free	2	5 / 10 Mbit/s

Tip: **family** is only for the family members who contributed to the Starlink (Part 13). Public customers can never get it unless you create a family login for them.

1 Upload the file

Files → drag `starling-profiles.rsc` from your PC into the Files window (or click **Upload** and choose it).

✔ **You should see:** The Files list shows `starling-profiles.rsc` at the top level (not inside a folder).

2 Import it

New Terminal → type:

```
/import file-name=starling-profiles.rsc
```

✔ **You should see:** Seven lines like “Profile 1hour (1h, 100 FCFA, 1 device(s)) ready” and “Profile family (free, never expires…) ready” and “Script file loaded and executed successfully”. **IP → Hotspot → User Profiles** lists the six profiles.

✖ **If not:** “no such file”: the file is not at the top level or the name is different; check **Files**. To try again, simply run the import again: it replaces the profiles.

Tip: Already imported an older version of this file? Upload the new one (replace) and run the import again. It updates the profiles in place and adds **family**; customers who are online keep working.

What each profile does (you do not need to change anything)

Setting	Value and meaning
Shared Users	**1**: one phone at a time per username (2 for 1month).
Rate Limit	**2M/5M**: 2 Mbit/s upload, 5 Mbit/s download per customer, so one person downloading a film does not slow everyone.
Idle Timeout	**none**: a customer is not logged out just for being idle.
Keepalive Timeout	**3h**: if a phone switches Wi-Fi off, it keeps its place for 3 hours, so the same username cannot be passed to a friend's phone.
Status Autorefresh	**1m**: the “time left” page refreshes every minute.
Scripts → On Login	At the first login: creates a timer (in **System → Scheduler**, named `exp-username`) and writes the start and end time in the user's Comment. When the timer ends: disconnects the phone and deletes the user.

Tip: To change a speed later: **IP → Hotspot → User Profiles** → double-click the profile → Rate Limit (e.g. `3M/8M`) → OK. To change a duration, edit the number after `interval=` in the profile's **Scripts → On Login** box (e.g. `1h`, `2d`, `7d`).

PART 12

Your Juthelink login page

Replace MikroTik's grey page with yours: plans, “Pay with MTN MoMo” (the payment code is filled in automatically for the plan chosen), WhatsApp, and the login form. The three files together are under 10 KB, tiny for the router.

Tip: Where are the files? In the kit zip `Starling-Router-Setup-Kit.zip`, folder `hotspot`: `login.html`, `alogin.html`, `status.html`. (In the project: `network/mikrotik/standalone/hotspot/`.) Your numbers are already inside: MoMo **677 358 806**, WhatsApp **673 727 995**. No editing needed.

1. **Unzip the kit on your PC**

   Right-click the zip → Extract. You get a folder with the PDF, `starling-profiles.rsc`, `ticket-maker.html` and the `hotspot` folder.

2. **Keep a copy of MikroTik's page**

   Winbox → **Files** → open the **hotspot** folder → right-click `hotspot/login.html` → rename to `hotspot/login-original.html`. (If your Winbox cannot rename files, skip this: the original pages can be restored any time with **IP → Hotspot → Server Profiles → Reset HTML**.)

3. **Upload the three pages into the hotspot folder**

   Drag `login.html`, `alogin.html` and `status.html` from the kit's `hotspot` folder onto the **hotspot** folder in the Files window (or click **Upload** while the hotspot folder is open). Replace existing files when asked.

✔ **You should see:** `hotspot/login.html`, `hotspot/alogin.html`, `hotspot/status.html` with today's date, **inside** the hotspot folder, exactly those names.

✖ **If not:** They appear at the top level: drag them onto the folder name, or remove and upload again. Close and reopen Files if the list does not refresh.

4. **Remove the logout page**

   Delete `hotspot/logout.html` (customers must not log out and pass their login to someone else). Leave every other file in the hotspot folder (rlogin.html, redirect.html, error.html, errors.txt, md5.js, css, img): the router uses some of them itself. To get MikroTik's original pages back at any time: **IP → Hotspot → Server Profiles → Reset HTML**.

5. **Test**

   On your phone (not bypassed): join the Wi-Fi → open `http://neverssl.com`.

✔ **You should see:** The blue **Juthelink** page: 1 Choose a plan, 2 Pay with MTN MoMo, 3 Send confirmation on WhatsApp, 4 Log in. Tap a plan: the yellow button and the code below it change (for 1 hour: `*126*9*677358806*100#`).

Juthelink

Fast satellite Wi-Fi. Choose a plan, pay with MTN MoMo, and we send you your login.

1 Choose a plan

100 F — 1 hour	150 F — 12 hours
200 F — 24 hours ✓	1 350 F — 1 week
2 500 F — 2 weeks	4 500 F — 1 month

Selected: 24 hours · 200 FCFA

2 Pay with MTN MoMo

Pay 200 FCFA with MTN MoMo

*126*9*677358806*200#

Tap the button: your phone opens the dialer with the code. Press call and enter your MoMo PIN. If the dialer does not open, dial the code above yourself.

3 Get your login on WhatsApp

Send payment confirmation

We check your payment and reply with your username and password, usually within a few minutes. WhatsApp uses your mobile data.

4 Log in

Username

Password

Connect

Family member? Log in with the username and password you were given (no payment needed).

Lost your login? Ask us on WhatsApp. If your time has not ended, we send it again.

▸ Can't connect, or asked to log in again and again?

Your time starts at your first login and ends automatically. One login works on one phone.

Login page

TIME LEFT

1h12m40s

Logged in as **k7m4qx**

Connected for	47m20s
Downloaded	412.7 MiB
Uploaded	38.2 MiB

Buy more time (WhatsApp)

Time left

PART 13

How selling works (MTN MoMo + WhatsApp)

There is no payment API yet, so **you** confirm each payment. The router does everything else: it starts the clock at the first login and disconnects the customer when the time is up.

#	The customer	You
1	Joins the Wi-Fi, the login page opens, chooses a plan.	—
2	Taps **Pay with MTN MoMo**: the phone dialer opens with `*126*9*677358806*AMOUNT#`. Presses call, enters the MoMo PIN.	Your MoMo phone receives the payment SMS.
3	Taps **Send payment confirmation**: WhatsApp opens with a ready message (plan + amount); adds their MoMo number and sends.	You check the MoMo SMS matches (amount and number).
4	—	Open **ticket-maker.html** on your PC → choose the plan → type the customer's phone → **Make ticket** → copy the command into Winbox **New Terminal** → Enter → copy the WhatsApp reply and send it.
5	Types the username and password in step 4 of the login page → **Connect**. The Time left page appears.	Done. The router ends their access automatically.

Tip: iPhones do not allow websites to start USSD codes. iPhone users dial the code shown under the yellow button themselves. Some Android “sign in to network” windows also block it: the customer can copy the code or dial it by hand.

Where do usernames come from?

You create them with **ticket-maker.html** (open it by double-clicking; it works offline on your PC). It makes a random username (6 letters/numbers, e.g. `k7m4qx`) and a 6-digit password, and gives you the exact Winbox command, for example:

```
/ip hotspot user add name=k7m4qx password=739215 profile=1hour comment="677123456"
```

The customer's phone number goes into the **comment**. At the first login the router adds the start and end time to that comment, so you can always see who, when and until when.

“I lost my login”

The login page has a **Lost your login?** link that opens WhatsApp with “I lost my Wi-Fi login. My phone number is …”. To find it: Winbox → **IP → Hotspot → Users → Find** → type the phone number. The matching line shows the username, password and when it ends. If it has not ended, send it again. If the user is gone, their time had ended (expired users are deleted automatically).

Tip: A customer who already logged in on their phone is usually let back in automatically (MAC cookie), even without typing the login again, until their time ends.

Family members (free access)

For family members who contributed to buying the Starlink: open **ticket-maker.html** → Plan: **Family member (free)** → type their name (and phone) → **Make ticket** → paste the command into Winbox → send them the WhatsApp message. Example command:

```
/ip hotspot user add name=marie766 password=261043 profile=family comment="FAMILY: Marie Ngono, 677123456"
```

- Free, never expires, up to **2 of their own devices**, faster speed (5/10 Mbit/s).
- Everyone can be told apart: in **IP → Hotspot → Users**, family logins have profile **family** and a comment starting with **FAMILY:**. Use **Find** → `FAMILY` to list them all.
- If a family login is shared with the public, change its password (double-click → Password) or click **✗ Disable**. Only you can create family logins.

Printed tickets for later

In ticket-maker, leave the phone empty and choose e.g. 20 tickets → paste the commands → **Print** the list and cut it. Each ticket's time starts at its first login, so they can wait in a drawer.

Check a ticket works

Make one 1-hour ticket for yourself and log in on your phone.

> **✔ You should see:** “You are connected”, then **Time left** counting down from about 1 hour. In Winbox: **IP → Hotspot → Active** shows the user; **Users** shows its comment “phone | Started … | ends …”; **System → Scheduler** shows `exp-username`.

PART 14

Tenda outdoor access point

The Tenda must only **broadcast the Wi-Fi** (“AP mode”). The MikroTik does everything else. If the Tenda acts as a router, all customers look like one phone and the hotspot cannot tell them apart.

14.1 Configure the Tenda on your desk first

1. **Wire it to your PC (not to the MikroTik yet)**
   Tenda cable → **PoE** port of the Tenda injector. Injector **LAN** port → your PC. Injector power → socket. Wait 1–2 minutes for the Tenda to start.

2. **Reset the Tenda**
   With power on, press and hold its reset button (or the button on the unit) for about 8 seconds, until the lights flash. Wait 1 minute.

3. **Give your PC an address in the Tenda's range**
   Look at the label on the Tenda for the default address (often `192.168.2.1`, user `admin`, password `admin`). On Windows: Settings → Network → Ethernet → IP assignment → Edit → **Manual**, IPv4 on, IP address `192.168.2.100`, Subnet mask `255.255.255.0`, Gateway empty → Save.

4. **Open the Tenda settings**
   Browser → `http://192.168.2.1` (or the address on the label) → log in. Set a new admin password if asked (write it in Part 0).

5. **Choose AP mode**
   Find **Working Mode / Operation Mode / Quick Setup** → choose **AP** (Access Point). **Not** Router, WISP, Client+AP or Repeater.

6. **Wi-Fi name and security**
   Wireless / Wi-Fi settings: SSID `Starling Wi-Fi` (exactly the same as the MikroTik), Security **None / Open**, Country **Cameroon** (if listed), Channel **1, 6 or 11** (choose a different one from the hAP lite if you keep both), Channel width **20 MHz**. If there is **Client Isolation / AP Isolation**, turn it **on**.

7. **LAN address and DHCP**
   LAN / Network settings: IP Address `10.5.50.2`, Subnet mask `255.255.255.0`, Gateway `10.5.50.1`, DNS `10.5.50.1`. **DHCP server: OFF** (if the option exists). Save. The Tenda reboots; your browser loses it. That is expected.

8. **Put your PC back to automatic**
   Windows: Ethernet → IP assignment → Edit → **Automatic (DHCP)** → Save.

14.2 Connect the Tenda to the MikroTik

MikroTik port 2 → cable → **Injector LAN** **Injector PoE** → long outdoor cable → **Tenda**

The injector also plugs into the power socket. Never plug the PoE port into the MikroTik or your PC.

1. **Plug in**
   Short cable from **MikroTik port 2** to the injector's **LAN** port. Keep the PoE port going to the Tenda.

2. **Check the Tenda is visible to the MikroTik**
   Winbox → **New Terminal**:

   ```
   /ping 10.5.50.2 count=4
   ```

✔ **You should see:** Four replies. From your PC (bypassed), `http://10.5.50.2` opens the Tenda settings.

✖ **If not:** No reply: check the cable is in port 2 and the injector's LAN port; redo 14.1 step 7 (address 10.5.50.2, gateway 10.5.50.1).

3. **Test through the Tenda**
   Stand near the Tenda (or turn off the hAP lite Wi-Fi for the test: **Wireless** → select wlan1 → **✗ Disable**). Phone → Starling Wi-Fi → open `http://neverssl.com`.

✔ **You should see:** The blue Starling login page. In Winbox, **IP → DHCP Server → Leases** shows your phone's own MAC address and a 10.5.50.x address.

✖ **If not:** The phone gets a 192.168.x.x address, or several phones show the same MAC in **IP → Hotspot → Hosts**: the Tenda is still in router mode or its DHCP is on. Redo 14.1 steps 5 and 7.

14.3 Mounting outdoors

- Mount it 3–6 m high on a pole or wall, facing where customers sit, antennas pointing up.
- Leave a “drip loop” (a U-shaped bend) in the cable below the Tenda so rain runs off, and close the cable cover tightly.
- Keep the injector indoors and dry.
- If you keep the hAP lite's own Wi-Fi on (indoors), use a different channel from the Tenda (e.g. 1 and 11). Same name is fine: phones pick the strongest.

PART 15

Final test checklist

Tick every line. Use a phone that is not bypassed (Part 9), with private Wi-Fi address turned off.

✓	Test	Expected
☐	Winbox → New Terminal → `/ping google.com count=4`	4 replies
☐	Phone joins **Starling Wi-Fi** near the Tenda	Connected, no password asked
☐	Open `http://neverssl.com`	Blue Starling login page
☐	Log in with a wrong password	Red message “invalid username or password”
☐	Log in with `test001` / `123456` (1hour)	“You are connected”, then Time left page
☐	Open YouTube / WhatsApp	Works
☐	A second phone tries the same username at the same time	Refused (“no more sessions are allowed”)
☐	Turn Wi-Fi off and on again on the first phone	Back online without typing (MAC cookie)
☐	Winbox **IP → Hotspot → Users**	test001 Comment shows Started … | ends …
☐	Wait 1 hour (or test with a `10m` profile)	Phone disconnected, login page appears again, test001 deleted
☐	Unplug the router power, plug back in	After 1 minute everything works again

Tip: Quick expiry test without waiting an hour: **IP → Hotspot → User Profiles → +** → Name `test10min`, Shared Users 1 → **Scripts** tab → paste the On Login script from `on-login-script.txt` and change `DURATION` to `10m` → OK. Create a user with this profile and log in. Delete the profile after the test.

Save your work (backup)

1. **Make a backup on the router and copy it to your PC**

   Files → Backup → Name `starling-working` → Backup. Then drag `starling-working.backup` from the Files window to your PC. Also in the terminal:

   ```
   /export file=starling-working
   ```

   and copy `starling-working.rsc` to your PC too (it is a readable text copy of all settings).

Tip: To go back to this working state one day: **Files** → select `starling-working.backup` → **Restore**.

PART 16

Daily use

I want to…	Where in Winbox
Sell a pass	Check the MoMo SMS → **ticket-maker.html** → paste the command in Winbox → send the WhatsApp reply (Part 13).
Give a family member free access	ticket-maker.html → **Family member (free)** → name → paste the command (Part 13).
See all family logins	**IP → Hotspot → Users → Find** → `FAMILY`.
Find a customer's login (lost login)	**IP → Hotspot → Users → Find** → type their phone number.
See who is online now	**IP → Hotspot → Active** (user, address, uptime, data).
See when a customer's time ends	**IP → Hotspot → Users** → Comment column (“Started … | ends …”), or **System → Scheduler** → `exp-username` → Next Run.
Disconnect a phone (e.g. customer changed phone)	**IP → Hotspot → Active** → select the line → **−** (Remove). The customer can log in again on the new phone; their time keeps running.
Stop a customer completely	**IP → Hotspot → Users** → select → **✗ Disable**, then remove them from Active.
Give someone free internet (staff)	**IP → Hotspot → IP Bindings → +**, their MAC, type bypassed.
Change prices on the login page	Edit login.html on your PC and upload it again into the hotspot folder (Part 12).
Check the router is healthy	Bottom bar of Winbox: CPU % and memory. CPU often above 70 % when busy means it is time for a bigger router.

Tell every customer

- **Turn off private / random Wi-Fi address** for Starling Wi-Fi. Android: Settings → Wi-Fi → gear → Privacy → **Use device MAC**. iPhone: Settings → Wi-Fi → ⓘ → turn off **Private Wi-Fi Address**. Otherwise the Wi-Fi keeps asking them to log in.
- Type the username and password exactly, with no spaces.
- One ticket works on one phone, and the time starts at the first login.

PART 17

Problems and fixes

Problem	Fix
Winbox cannot find the router	Use the MAC address. Replug the yellow cable. Try another port. Last resort: hardware reset (unplug power, hold reset, plug power, keep holding about 5 s until the light flashes, release), then start again at Part 2.
DHCP Client on ether1 not “bound”	Blue cable in port 1? Starlink app Online? Restart the Starlink router.
Customers get internet without logging in	FastTrack rule active (Part 8 step 2), or the device is bypassed (Part 9), or the hotspot server hotspot1 is disabled.
Grey MikroTik page instead of the blue one	Files not inside the hotspot folder or wrong name (Part 12).
No login page pops up	Open `http://neverssl.com` or `http://login.starling`. HTTPS sites (Google, Facebook) cannot show the login page; that is normal.
“invalid username or password”	Extra space or capital letter. Check the user in IP → Hotspot → Users. The user may have expired (deleted).
“no more sessions are allowed for user”	Another phone is using this ticket (or the same phone with a new random MAC). Remove it from Active, and ask the customer to turn off private Wi-Fi address.
Customer asked to log in again and again	Private / random Wi-Fi address is on. See Part 16.
Phones get 192.168.x.x addresses	Tenda DHCP is on or Tenda is in router mode (Part 14.1 steps 5 and 7).
All phones show the same MAC in IP → Hotspot → Hosts	Tenda is in router mode. Put it in AP mode.
Time ended too early or too late	Clock wrong: System → Clock and System → SNTP Client (Part 3 step 3). Check internet works.
Profiles import fails	The file must be at the top level in Files. Run the import again.
Everything is slow	Check Starlink speed in the Starlink app. Lower the per-customer speed in the profiles. Check router CPU.

APPENDIX A

All commands in one place

Exactly what the clicks above do, for the terminal. Run them in order, one block at a time, after Part 2 (start clean). Replace the values in CAPITALS first.

```
# Part 3 - password, name, clock
/password old-password="" new-password="YOUR-PASSWORD" confirm-new-password="YOUR-PASSWORD"
/system identity set name=starling-hotspot
/system clock set time-zone-name=Africa/Douala
/system ntp client set enabled=yes primary-ntp=162.159.200.1 secondary-ntp=162.159.200.123

# Part 4 - internet from Starlink on port 1
/ip dhcp-client add interface=ether1 use-peer-dns=no add-default-route=yes disabled=no
comment="Internet from Starlink"
/ip dns set servers=1.1.1.1,8.8.8.8 allow-remote-requests=yes
/ping google.com count=4

# Part 6 - customer bridge
/interface bridge add name=bridge-hotspot comment="Customers"
/interface bridge port add bridge=bridge-hotspot interface=ether2
/interface bridge port add bridge=bridge-hotspot interface=ether3
/interface bridge port add bridge=bridge-hotspot interface=ether4
/interface bridge port add bridge=bridge-hotspot interface=wlan1
/ip address add address=10.5.50.1/24 interface=bridge-hotspot comment="Customer gateway"

# Part 7 - the hAP lite's own open Wi-Fi
/interface wireless security-profiles set [find default=yes] mode=none
/interface wireless set wlan1 mode=ap-bridge band=2ghz-b/g/n channel-width=20mhz frequency=auto
ssid="Starling Wi-Fi" security-profile=default wireless-protocol=802.11 disabled=no

# Part 8 - hotspot (what the Hotspot Setup wizard creates)
/ip pool add name=hs-pool ranges=10.5.50.10-10.5.50.250
/ip dhcp-server add name=dhcp-hotspot interface=bridge-hotspot address-pool=hs-pool lease-
time=1h disabled=no
/ip dhcp-server network add address=10.5.50.0/24 gateway=10.5.50.1 dns-server=10.5.50.1
/ip hotspot profile add name=hsprof1 hotspot-address=10.5.50.1 dns-name=login.starling html-
directory=hotspot smtp-server=0.0.0.0
/ip hotspot add name=hotspot1 interface=bridge-hotspot address-pool=hs-pool profile=hsprof1
disabled=no
/ip hotspot user add name=admin password="YOUR-TEST-PASSWORD"
/ip firewall nat add chain=srcnat src-address=10.5.50.0/24 action=masquerade
comment="masquerade hotspot network"
/ip firewall filter disable [find action=fasttrack-connection]

# Part 9 - your own devices skip the login
/ip hotspot ip-binding add mac-address=YOUR:PC:MAC:AA:BB:CC type=bypassed comment="Owner PC"

# Part 10 - hotspot settings
/ip hotspot set [find] addresses-per-mac=1
/ip hotspot profile set hsprof1 login-by=http-chap,http-pap,mac-cookie,cookie

# Part 11 - packages (upload starling-profiles.rsc to Files first)
/import file-name=starling-profiles.rsc

# Part 13 - a customer
/ip hotspot user add name=frank482 password=739215 profile=24hours

# Backup
/export file=starling-working
```

Tip: Every command on this page was tested on RouterOS 6.48 before this guide was written. When you create the hotspot with the terminal (Part 8) instead of the wizard, the `hotspot` folder with MikroTik's default pages is created automatically, exactly as with the wizard.

Phase 2 (later): the Starling website and payments connect to this same router through a secure tunnel. It uses its own profiles and users and never touches the ones you create here. See `docs/network-integration-guide.md` Part 8.